

SCOPE OF APPLICATION All Project/Engineering		SHT/SHTS 1 / 26
Responsibility: 제어SW플랫폼1팀	AUTOSAR Nm Manual	DOC. NO
AUTOSAR Nm User Manual		

Document Change Histroy				
Date (YYYY-MM-DD)	Ver.	Editor	Chap	내용(개정 전 -> 개정 후)
2016-04-08	2.1.1	Kyungtae Kim		• Nm UM 분리
2019-01-07	2.2.0	JeongSu Lim	4.2 4.3 5 6.3 8.2	• Scope of the Release 변경. • Change Log 추가. • Configuration Guide 수정. • API Functions 추가. • Use Case별 설정 Guide 추가.
2019-01-23	2.2.1	JeongSu Lim	4.2 4.3	• Scope of the Release 변경 • Change Log 추가
2020-11-30	2.3.0.0	Hyungmin Shin	4.2 4.3	• Scope of the Release 변경 • Change Log 추가

Last Edition Date: 30 Nov, 2020	File Name AUTRON_AUTOSAR_Nm_UM.pdf	Creation <i>Hyungmin Shin</i> 2020/11/30	Check <i>JongSun Lim</i> 2020/11/30	Approval <i>JongHyun Baek</i> 2020/11/30
Document Management System				

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1. Overview

Autosar 표준 SRS/SWS 를 기반으로 작성 되었으며, 모듈 사용시 보다 자세한 기능적인 설명이 필요한 경우, 아래 Reference 문서를 참고한다

설정관련 Category 의 해석은 다음과 같다.

- Changeable (C) : User 에 의해서 설정 가능한 항목
- Fixed (F) : User 에 의한 변경이 불가한 항목
- NotSupported (N) : 사용되지 않는 항목

2. Reference

Sl. No.	Title	Version
1	Specification of Network Management Interface	3.0.0
2		
3		

3. AUTOSAR System

3.1 CAN Communication Stack

현대오토론 AUTOSAR 플랫폼에서 CAN Communication Stack 은 아래 세부 모듈로 구성된다.

- CanIf : CAN 메시지 송수신을 담당
- PduR : 통신 모듈 간 PDU 전달을 담당
- IpduM : Multiplexed PDU 송수신을 담당
- CanTp : Transport Protocol 기반의 대용량 데이터 송수신을 담당
- CanSM : CAN 통신 채널의 상태 제어 및 Bus-Off 처리를 담당
- CanTrcv : CAN 트랜시버 하드웨어 제어를 담당
- OsekNm/Nm : CAN 통신 채널의 SLEEP 진입 동기화를 담당
- CanCM : 배터리 전압 및 HKMC 사양을 기준으로 CAN 통신 활성화 및 비활성화 제어를 담당

3.2 Nm Module

Nm 모듈은 통신 채널의 Sleep, Wake-up 제어를 위해 아래와 같은 동작을 수행한다.

- 하위 <BusNm> 모듈의 상태 전환을 ComM 에 통지
- ComM 모듈의 통신 모드 전환 요청을 하위 <BusNm> 모듈에게 통지

4. Product Release Notes

4.1 Overview

이 Chapter 에서는, 현대오토론 Nm 모듈에 대한 release 관련 내용을 제공하는데 목적이 있으며, Nm Software product release version 에 대한 제한사항 및 특이사항을 기술하고 있다.

4.2 Scope of the Release

이 문서에 대한 모든 내용은, 다음의 현대오토론 Nm 모듈에 한정한다.

Module name	AUTOSAR version	SWS version	Module version
Nm	4.0.3	3.0.0	2.3.0

※ Module version 은 각 모듈의 BswModule Description(Bswmd)파일의 Sw version 을 의미한다.

4.3 Change Log

4.3.1 Version 2.3.0.0

기능개선

■ Misra-C Verification

원인	Misra-C Verification
동작영향	없음
설정영향	없음

ASW 조치 필요 사항	없음
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신규기능

- PduR M:N Configuration 시에도 동작 가능하도록 기능 개선

원인	PduR M:N Configuration 시에도 동작 가능하도록 기능 개선
동작영향	없음
설정영향	없음
ASW 조치 필요 사항	없음

4.3.2 Version 2.2.1

오류수정

- Tx ComSignal 이 0 개 일 때 Com_SendSignal 호출 코드로 인한 Compile Error 해결

원인	Tx ComSignal이 0개 일 때 Com_SendSignal 호출 코드로 인한 Compile Error 해결
동작영향	없음
설정영향	없음
ASW 조치 필요 사항	없음

4.3.3 Version 2.2.0

기능개선

- NmStateChangeIndUserCallout 기능 추가

원인	NmStatechangeIndUserCallout 기능 추가
동작영향	없음
설정영향	없음
ASW 조치 필요 사항	없음

- NmPduRxIndUserCallout 기능 추가

원인	NmPduRxIndUserCallout 기능 추가
동작영향	없음
설정영향	없음
ASW 조치 필요 사항	없음

4.4 Module Release Notes

4.4.1 Limitations

- Coordination Support

하나의 클러스터에 속한 서로 다른 네트워크의 SLEEP 진입을 동기화하는 기능. 클러스터에 속한 모든 네트워크에 CanNm 모듈이 적용되어야 하므로 현재 Odin 플랫폼은 해당 기능을 지원하지 않는다.

- Passive Mode Support

Passve 모드로 설정된 네트워크 노드는 NM Message 를 송신하지 않고 네트워크 전체의 SLEEP 및 Wake-up 상태에 따라 동작한다. 현재 Odin 플랫폼은 해당 기능을 지원하지 않는다.

- State Report Signal

송신 Signal 지정하여 현재의 NM 상태 정보를 외부로 전송 가능하다. 현재 Odin 플랫폼은 해당 기능을 지원하지 않는다.

- Car Wake Up Support

Nm 메시지 상의 CWU 비트 값을 이용한 Wake-up 기능이며 CanNm 모듈과 연동이 필요하여 현재 Odin 플랫폼은 해당 기능을 지원하지 않는다.

4.4.2 Deviations

None

5. Configuration Guide

5.1 NmGlobalFeatures

Parameter Name	Value	Category
NmBusSynchronizationEnabled	False	N
NmCarWakeUpRxEnabled	False	N
¹⁾ NmComControlEnabled	True	F
²⁾ NmComUserDataSupport	True	F
NmCoordinatorSupportEnabled	False	N
³⁾ NmNodeDetectionEnabled	False	C
⁴⁾ NmNodeIdEnabled	False	C
⁵⁾ NmPduRxIndicationEnabled	False	C
NmRemoteSleepEnabled	False	N
NmRepeatMsgIndEnabled	False	N
⁶⁾ NmStateChangeIndEnabled	False	C
⁷⁾ NmUserDataEnabled	True	F
NmCoordinatorSyncSupport	False	N
NmGlobalCoordinatorTime	0	N
NmCarWakeUpCallback		N
⁸⁾ NmHeaderFileInclusion		C

- 1) NmComControlEnabled :
NM PDU 송수신 제어를 위한 기능을 제공한다.(Nm_EnableCommunication, Nm_DisableCommunication)
- 2) NmComUserDataSupport:
NM PDU 의 UserData 에 대한 쓰기 방식을 결정한다.(True : Com 모듈을 통한 UserData 접근, False : Nm_SetUserData API 를 통한 접근)
- 3) NmNodeDetectionEnabled:
RepeatMessage Request 처리를 위한 기능을 제공한다.
Nm_RepeatMessageRequest
- 4) NmNodeIdEnabled:
NodeIdIdentifier 처리를 위한 기능을 제공한다.
- 5) NmPduRxIndicationEnabled
Pdu Rx Indication 처리를 위한 기능을 제공한다.(Nm_PduRxIndication)
- 6) NmStateChangeIndEnabled
State Change Indication 처리를 위한 기능을 제공한다.(Nm_StateChangeNotification)
- 7) NmUserDataEnabled

NM PDU 의 User Data 처리를 위한 기능을 제공한다.(Nm_SetUserData, Nm_GetUserData)

8) NmHeaderFileInclusion

User Defined Header File 을 Include 하는 기능을 제공한다.

5.2 NmGlobalConstants

Parameter Name	Value	Category
¹⁾ NmNumberOfChannels	-	Fixed

1) NmNumberOfChannels

NmChannel 의 개수를 지정한다.

5.3 NmGlobalProperties

Parameter Name	Value	Category
¹⁾ NmCycleMainFunction	0.01	F
²⁾ NmDevErrorDetect	False	C
³⁾ NmVersionInfoApi	False	C

1) NmCycleMainFunction

Nm_MainFunction 의 주기를 지정한다.

2) NmDevErrorDetect

Development Error 를 Detect 하는 기능을 제공한다.

3) NmVersionInfoApi

VersionInfo 를 제공하기 위한 기능을 제공한다.(Nm_GetVersionInfo)

5.4 NmChannelConfig

Parameter Name	Value	Category
NmActiveCoordinator	False	N
¹⁾ NmChannelId	-	F
NmChannelSleepMaster	False	N
NmCoordClusterIndex	-	N
NmPassiveModeEnabled	False	N
NmShutdownDelayTimer	0	N
NmStateReportEnabled	False	N
NmSynchronizingNetwork	False	N
²⁾ NmComMChannelRef	-	F
NmStateReportSignalRef	-	N
³⁾ NmStateChangeIndUserCallout		C
⁴⁾ NmPduRxIndUserCallout		C

1) NmChannelId

NmChannel 의 Id 를 ComMChannel 의 Id 와 동일하게 지정한다.

2) NmComMChannelRef

NmChannel 에 대응하는 ComMChannel 을 지정한다.

3) NmStateChangeIndUserCallout

Nm_StateChangeNotification 에서 호출할 User Callout 함수를 지정한다.

OsekNm 과 같은 GenericBusNm Type 은 지원하지 않는다.

4) NmPduRxIndUserCallout

Nm_PduRxIndication 에서 호출할 User Callout 함수를 지정한다.

6. Application Programming Interface (API)

6.1 Type Definitions

None

6.2 Macro Constants

None

6.3 Functions

6.3.1 Nm_GetNodeIdentifier

Function Name	Nm_GetNodeIdentifier	
Syntax:	Std_ReturnType Nm_GetNodeIdentifier(NetworkHandleType NetworkHandle, uint8* nmNodeIDPtr)	
Service ID	0x0a	
Sync/Async	Synchronous	
Reentrancy	Non-Reentrant	
Parameters (In)	NetworkHandle	Identification of the NM-channel
Parameters (Inout)	None	
Parameters (Out)	nmNodeIDPtr	Pointer where node identifier out of the last successfully received NM-message shall be copied to
Return Value	Std_ReturnType	E_OK: No error E_NOT_OK: Getting of the node identifier out of the last received NM-message has failed
Description	Get node identifier out of the last successfully received NM-message.	
Preconditions	Nm_Init() should be initialized.	
Configuration Dependency	Nm/NmGlobalConfig/NmGlobalFeatures/NodeIDEnabled == true.	

6.3.2 Nm_GetPduData

Function Name	Nm_GetPduData	
Syntax:	Std_ReturnType Nm_GetPduData(NetworkHandleType NetworkHandle, uint8* nmPduData)	
Service ID	0x08	
Sync/Async	Synchronous	
Reentrancy	Non-Reentrant	
Parameters (In)	NetworkHandle	Identification of the NM-channel
Parameters (Inout)	None	
Parameters (Out)	nmPduData	Pointer where NM PDU shall be copied to.
Return Value	Std_ReturnType	E_OK: No error E_NOT_OK: Getting of NM PDU data has failed
Description	Get the whole PDU data out of the most recently received NM message.	
Preconditions	Nm_Init() should be initialized.	
Configuration	Nm/NmGlobalConfig/NmGlobalFeatures/NodeIDEnabled == true	

Dependency

Or

Nm/NmGlobalConfig/NmGlobalFeatures/UserDataEnabled == true

7. Generator

7.1 Generator Option

Option	Description
-V	현재 사용 중인 모듈의 버전을 표시한다.
-O	Output 이 생성될 위치를 지정한다.

7.2 Generator Message

7.2.1 Error Messages

ERR029001: Unexpected Error Found. Please contact AUTRON AUTOSAR Support System.

This error occurs, if the number of fields is not same in the structure that is to be generated in the C Source file. Contact AUTRON.

ERR029002: Unexpected Error Found. This error may be due to the incorrect configuration of the element(s) 'Element Name'. If the error is not resolved, then please contact AUTRON.

This error may occur due to incorrect configuration of the Parameter Name/ Container Name provided in the error message. If the error is not resolved, then contact AUTRON. For more information refer [Section 4](#).

ERR029003: 'Component Name' Component is not present in the input file(s).

This error occurs, if any one of Nm or ComM or Com component is not present in any of the input ECU Configuration Description File(s).

ERR029004: The Reference path is empty for the parameter 'NmComMChannelRef' in the container 'NmChannelConfig', having short name 'NmChannelConfig0'.

This error occurs, if no reference path is configured for the parameter NmComMChannelRef in the container NmChannelConfig.

ERR029005: Parameter 'Parameter Name' in the container 'Container Name' should be configured.

This error occurs, if value of any of the mandatory parameters mentioned in the below table are not configured.

Container Name	Parameter Name
BSW-IMPLEMENTATION	AR-RELEASE-VERSION
	VENDOR-ID

Container Name	Parameter Name
	SW-VERSION
BSW-MODULE-DESCRIPTION	MODULE-ID
NmChannelConfig	NmChannelId
	NmPassiveModeEnabled
	NmStateReportEnabled
	NmComMChannelRef
NmGenericBusNmConfig	NmGenericBusNmPrefix
NmStandardBusNmConfig	NmStandardBusType
NmGlobalConstants	NmNumberOfChannels
NmGlobalFeatures	NmBusSynchronizationEnabled
	NmCarWakeUpRxEnabled
	NmComControlEnabled
	NmComUserDataSupport
	NmCoordinatorSupportEnabled
	NmNodeDetectionEnabled
	NmNodeIdEnabled
	NmPduRxIndicationEnabled
	NmRemoteSleepIndEnabled
	NmRepeatMsgIndEnabled
	NmStateChangeIndEnabled
	NmUserDataEnabled
	NmCoordinatorSyncSupport
NmGlobalProperties	NmCycletimeMainFunction
	NmDevErrorDetect
	NmVersionInfoApi

ERR029006: The value configured for the parameter 'Parameter Name' in the container 'Container Name' should follow the pattern: <Pattern>.

This error occurs, when the parameter 'Parameter Name' is not configured as per the pattern.

Parameter Name	Container Name	Pattern	Example
AR-RELEASE-VERSION	BSW-IMPLEMENTATION	4.[0-9]+.[0-9]+	4.0.3
SW-VERSION		1.[0-9]+.[0-9]+	1.0.0
NmGenericBusNmPrefix	NmGenericBusNmConfig	[a-zA-Z][a-zA-Z0-9W_]*	GenericNm

ERR029013: The reference path <Reference Path> provided for the parameter 'NmComMChannelRef' in the container 'NmChannelConfig', having short name <Short Name> is incorrect.

This error occurs, if the reference path provided for the parameter NmComMChannelRef in the

container NmChannelConfig is incorrect.

ERR029021: Number of Channels configured should be equal to the value configured for the parameter 'NmNumberOfChannels' in the container 'NmGlobalConstants'.

This error occurs, if the number of channels configured is not equal to the value configured for the parameter 'NmNumberOfChannels' in the container NmGlobalConstants.

ERR029022: Value of the parameter 'Parameter Name' in the container 'Container Name' should not be configured as <0>.

This error occurs, if value of the below mentioned parameters are configured as zero.

Parameter Name	Container Name
NmCycletimeMainFunction	NmGlobalProperties
NmShutdownDelayTimer	NmChannelConfig

ERR029023: Parameter 'NmCarWakeUpCallback' in the container 'NmGlobalFeatures' should be configured, since value of the parameter 'NmCarWakeUpRxEnabled' in the container 'NmGlobalFeatures' is configured as <true/1>.

This error occurs, if parameter NmCarWakeUpCallback is not configured, when the value of the parameter NmCarWakeUpRxEnabled is configured as <true/1> in the container NmGlobalFeatures.

ERR029024: Calculated number of ticks = INT (NmShutdownDelayTimer / NmCycletimeMainFunction) should not be zero.

This error occurs, if calculated number of ticks for parameter NmShutdownDelayTimer divided by NmCycletimeMainFunction results in a value of zero.

ERR029051: Number of Channels configured should not be <1> when the value configured for the parameter 'NmNumberOfChannels' in the container 'NmGlobalConstants' is greater than <1>.

This error occurs, if Number of Channels configured is <1> when the value of the parameter 'NmNumberOfChannels' in the container NmGlobalConstants is configured as <1>.

ERR029052: Value of the parameter 'NmStateReportEnabled' should not be configured as <true/1>, when the value of the parameters 'NmStateChangeIndEnabled' and '<Bus>NmComUserDataSupport' are configured as <false/0>.

This error occurs, if the value of the parameter NmStateReportEnabled is configured as <true/1>, when the value of the parameters NmStateChangeIndEnabled and <Can/Fr/Lin/Udp>NmComUserDataSupport are configured as <false/0>.

ERR029053: Value of the parameter 'Dependent Parameter Name' in the container 'NmGlobalFeatures' should be configured as <false/0>, when the value of the parameter

‘Parameter Name’ in the container ‘NmChannelConfig’ is configured as <true/1>.

This error occurs, if value of the below mentioned dependent parameter is configured as <true/1> when the value of the dependent parameter is configured as <true/1> in the container NmGlobalFeatures.

Dependent Parameter Name	Parameter Name
NmRemoteSleepIndEnabled	NmPassiveModeEnabled
NmBusSynchronizationEnabled	

ERR029054: The parameter ‘Parameter Name’ in the container ‘NmChannelConfig’ should be unique.

This error occurs, if any of the below mentioned parameters are not unique in the container NmChannelConfig.

Parameter Name	Container Name
NmChannelId	NmChannelConfig
NmComMChannelRef	

ERR029055: The reference path <Reference Path> provided for the parameter ‘NmStateReportSignalRef’ in the container ‘NmChannelConfig’, having short name <Short Name> is incorrect.

This error occurs, if the reference path configured for the parameter NmStateReportSignalRef in the container NmChannelConfig is incorrect when the value of the parameter NmStateReportEnabled is configured as <true/1>.

ERR029056: Value of the parameter ‘NmChannelId’ in the container ‘NmChannelConfig’ should be equal to the value of the parameter ‘ComMChannelId’ in the container ‘ComMChannel’.

This error occurs, if value of parameter NmChannelId in the container NmChannelConfig is not same as value of the parameter ComMChannelId collected through the reference parameter NmComMChannelRef.

ERR029057: Parameter ‘NmStateReportSignalRef’ the container ‘NmChannelConfig’ should be configured, since value of the parameter ‘NmStateReportEnabled’ in the container ‘NmChannelConfig’ is configured as <true/1>.

This error occurs, if the reference path is not configured for the parameter NmStateReportSignalRef in the container NmChannelConfig when the value of the parameter NmStateReportEnabled is configured as <true/1>.

ERR029058: Parameter ‘Parameter Name’ in the container ‘Container Name’ should be

configured, when the value of the parameter 'NmCoordClusterIndex' in the container 'NmChannelConfig' is configured.

This error occurs, if the parameter 'Parameter Name' in the container 'Container Name' is not configured, when the value of the parameter NmCoordClusterIndex in the container NmChannelConfig is configured.

Parameter Name	Container Name
NmChannelSleepMaster	NmChannelConfig
NmShutdownDelayTimer	
NmSynchronizingNetwork	

ERR029100: The number of NmGlobalFeatures in Nm shall be 1.

This error occurs, If the NmGlobalFeatures count is 0 or greater than 1, NmGlobalFeatures count shall be 1.

ERR029101: The number of NmGlobalConstants in Nm shall be 1.

This error occurs, If the NmGlobalConstants count is 0 or greater than 1, NmGlobalConstants count shall be 1.

ERR029102: The number of NmGlobalProperties in Nm shall be 1.

This error occurs, If the NmGlobalProperties count is 0 or greater than 1, NmGlobalProperties count shall be 1.

ERR029103: NmGlobalConfig in Nm is not set. NmGlobalConfig in Nm shall be set.

This error occurs, If the NmGlobalConfig count is 0 or greater than 1, NmGlobalConfig count shall be 1.

ERR029104: NmNumberOfChannels in NmGlobalConstants shall be greater than or equal to the maximum value of NmChannelConfig count.

This error occurs, NmNumberOfChannels in NmGlobalConstants shall be greater than or equal to the NmChannelConfig count.

ERR029105: NmBusSynchronizationEnabled in NmGlobalFeatures is not set. NmBusSynchronizationEnabled in NmGlobalFeatures shall be set.

This error occurs, NmBusSynchronizationEnabled in NmGlobalFeatures shall be set as True or False.

ERR029106: NmCarWakeUpRxEnabled in NmGlobalFeatures is not set. NmCarWakeUpRxEnabled in NmGlobalFeatures shall be set.

This error occurs, NmCarWakeUpRxEnabled in NmGlobalFeatures shall be set as True or False.

ERR029107: NmComControlEnabled in NmGlobalFeatures is not set. NmComControlEnabled in NmGlobalFeatures shall be set.

This error occurs, NmComControlEnabled in NmGlobalFeatures shall be set as True or False.

ERR029108: NmComUserDataSupport in NmGlobalFeatures is not set. NmComUserDataSupport in NmGlobalFeatures shall be set.

This error occurs, NmComUserDataSupport in NmGlobalFeatures shall be set as True or False.

ERR029109: NmCoordinatorSupportEnabled in NmGlobalFeatures is not set. NmCoordinatorSupportEnabled in NmGlobalFeatures shall be set.

This error occurs, NmCoordinatorSupportEnabled in NmGlobalFeatures shall be set as True or False.

ERR029110: NmNodeDetectionEnabled in NmGlobalFeatures is not set. NmNodeDetectionEnabled in NmGlobalFeatures shall be set.

This error occurs, NmNodeDetectionEnabled in NmGlobalFeatures shall be set as True or False.

ERR029111: NmNodeIdEnabled in NmGlobalFeatures is not set. NmNodeIdEnabled in NmGlobalFeatures shall be set.

This error occurs, NmNodeIdEnabled in NmGlobalFeatures shall be set as True or False.

ERR029112: NmPduRxIndicationEnabled in NmGlobalFeatures is not set. NmPduRxIndicationEnabled in NmGlobalFeatures shall be set.

This error occurs, NmPduRxIndicationEnabled in NmGlobalFeatures shall be set as True or False.

ERR029113: NmRemoteSleepIndEnabled in NmGlobalFeatures is not set. NmRemoteSleepIndEnabled in NmGlobalFeatures shall be set.

This error occurs, NmRemoteSleepIndEnabled in NmGlobalFeatures shall be set as True or False.

ERR029114: NmRepeatMsgIndEnabled in NmGlobalFeatures is not set. NmRepeatMsgIndEnabled in NmGlobalFeatures shall be set.

This error occurs, NmRepeatMsgIndEnabled in NmGlobalFeatures shall be set as True or False.

ERR029115: NmStateChangeIndEnabled in NmGlobalFeatures is not set. NmStateChangeIndEnabled in NmGlobalFeatures shall be set.

This error occurs, NmStateChangeIndEnabled in NmGlobalFeatures shall be set as True or False.

ERR029116: NmUserDataEnabled in NmGlobalFeatures is not set. NmUserDataEnabled in NmGlobalFeatures shall be set.

This error occurs, NmUserDataEnabled in NmGlobalFeatures shall be set as True or False.

ERR029117: NmCoordinatorSyncSupport in NmGlobalFeatures is not set. NmCoordinatorSyncSupport in NmGlobalFeatures shall be set.

This error occurs, NmCoordinatorSyncSupport in NmGlobalFeatures shall be set as True or False.

ERR029118: NmCycleTimeMainFunction in NmGlobalProperties is not set. NmCycleTimeMainFunction in NmGlobalProperties shall be set.

This error occurs, NmCycleTimeMainFunction in NmGlobalProperties shall be set.

ERR029119: NmDevErrorDetect in NmGlobalProperties is not set. NmDevErrorDetect in NmGlobalProperties shall be set.

This error occurs, NmDevErrorDetect in NmGlobalProperties shall be set as True or False.

ERR029120: NmVersionInfoApi in NmGlobalProperties is not set. NmVersionInfoApi in NmGlobalProperties shall be set.

This error occurs, NmVersionInfoApi in NmGlobalProperties shall be set as True or False.

ERR029121: NmChannelId in NmChannelConfig is not set. NmChannelId in NmChannelConfig shall be set.

This error occurs, NmChannelId in NmChannelConfig shall be set.

ERR029122: NmChannelSleepMaster in NmChannelConfig is not set. NmChannelSleepMaster in NmChannelConfig shall be set.

This error occurs, NmChannelSleepMaster in NmChannelConfig shall be set as True or False.

ERR029123: NmPassiveModeEnabled in NmChannelConfig is not set. NmPassiveModeEnabled in NmChannelConfig shall be set.

This error occurs, NmPassiveModeEnabled in NmChannelConfig shall be set as True or False.

ERR029124: NmStateReportEnabled in NmChannelConfig is not set. NmStateReportEnabled in NmChannelConfig shall be set.

This error occurs, NmStateReportEnabled in NmChannelConfig shall be set as True or False.

ERR029125: NmSynchronizingNetwork in NmChannelConfig is not set. NmSynchronizingNetwork in NmChannelConfig shall be set.

This error occurs, NmSynchronizingNetwork in NmChannelConfig shall be set as True or False.

ERR029126: NmComMChannelRef in NmChannelConfig is not set. NmComMChannelRef in NmChannelConfig shall be set.

This error occurs, NmComMChannelRef in NmChannelConfig shall be set.

ERR029127: NmShutdownDelayTimer in NmChannelConfig is not set. NmShutdownDelayTimer in NmChannelConfig shall be set.

This error occurs, NmShutdownDelayTimer in NmChannelConfig shall be set.

ERR029128: NmChannelConfig in Nm is not set. NmChannelConfig in Nm shall be set.

This error occurs, NmChannelConfig in Nm shall be set.

ERR029129: NmBusType in NmChannelConfig is not set. NmBusType in NmChannelConfig shall be set.

This error occurs, NmBusType in NmChannelConfig shall be set.

ERR029130: NmGenericBusNmConfig in NmBusType and NmStandardBusNmConfig in NmBusType shall be exclusively set.

This error occurs, NmGenericBusNmConfig in NmBusType and NmStandardBusNmConfig in NmBusType shall be exclusively set.

ERR029131: NmGenericBusNmPrefix in NmGenericBusNmConfig is not set. NmGenericBusNmPrefix in NmGenericBusNmConfig shall be set.

This error occurs, NmGenericBusNmPrefix in NmGenericBusNmConfig shall be set.

ERR029132: NmStandardBusType in NmStandardBusNmConfig is not set. NmStandardBusType in NmStandardBusNmConfig shall be set.

This error occurs, NmStandardBusType in NmStandardBusNmConfig shall be set.

ERR029133: NmStateChangeIndUserCallout in NmChannelConfig is set even though NmBusType

of NmChannel is GenericBusNm. If NmBusType of NmChannel is GenericBusNm, NmStateChangeIndUserCallout in NmChannelConfig shall not be set.

This error occurs, NmStateChangeIndUserCallout in NmChannelConfig shall not be set.

7.2.2 Warning Messages

7.2.3 Information Messages

INF029003: Parameter 'Parameter Name' in the container 'Container Name' is configured, when value of the parameter 'Parameter Name1' in the container 'Container Name1' is configured as <false/0>.

This information occurs, if the below mentioned parameters 'Parameter Name' are configured when value of the parameter 'Parameter Name1' is configured as <false/0>.

Parameter Name	Container Name	Parameter Name1	Container Name1
NmCycleTimeMainFunction	NmGlobalProperties	NmCoordinatorSupportEnabled	NmGlobalFeatures
NmGlobalCoordinatorTime	NmGlobalFeatures	NmBusSynchronizationEnabled	
NmNodeDetectionEnabled		NmNodeIdEnabled	
NmActiveCoordinator	NmChannelConfig	NmBusSynchronizationEnabled	

INF029015: AUTOSAR Release version <version> configured for the parameter 'AR-RELEASE-VERSION' in provided MDT file is not correct. AUTOSAR Release version should be one of the following: 4.0.3.

This information occurs, if the value of the element AR-RELEASE-VERSION present in the BSW Module Description template is configured other than 4.0.3.

INF029051: Value of the parameter 'NmStateReportSignalRef' in the container 'NmChannelConfig' is ignored, since value of the parameter 'NmStateReportEnabled' in the container 'NmChannelConfig' is configured as <false/0>.

This information occurs, if value of the parameter NmStateReportEnabled in the container NmChannelConfig is configured as <false/0>.

INF029052: The parameter 'NmCoordinatorSupportEnabled' in the container 'NmGlobalFeatures' should be configured as <false/0>, when the value of the parameter 'NmNumberOfChannels' in the container 'NmGlobalConstants' is configured as <1> and the value of the parameter 'NmRemoteSleepIndEnabled' in the container 'NmGlobalFeatures' is configured as <false/0> hence Generation Tool will reset the value of the parameter 'NmCoordinatorSupportEnabled' as <false/0>.

This information occurs, if value of the parameter 'NmCoordinatorSupportEnabled' in the container 'NmGlobalFeatures' is configured as <true/1>, when the value of the parameter 'NmNumberOfChannels' in the container 'NmGlobalConstants' is configured as <1> and the value of

the parameter 'NmRemoteSleepIndEnabled' in the container 'NmGlobalFeatures' is configured as <false/0> hence Generation Tool will reset the value of the parameter 'NmCoordinatorSupportEnabled' as <false/0>.

INF029054: Value of the parameter 'Parameter Name' in the container 'NmChannelConfig' is ignored, since value of the parameter 'Parameter Name1' in the container 'NmChannelConfig' is not configured.

This information occurs, if the below mentioned parameter NmCoordClusterIndex is not configured.

Parameter Name	Parameter Name1
NmChannelSleepMaster	NmCoordClusterIndex
NmShutdownDelayTimer	
NmSynchronizingNetwork	

8. Appendix

8.1 기능별 설정 Guide

8.2 Use Case 별 설정 Guide

8.2.1 Nm State 에 따른 제어가 필요할 경우

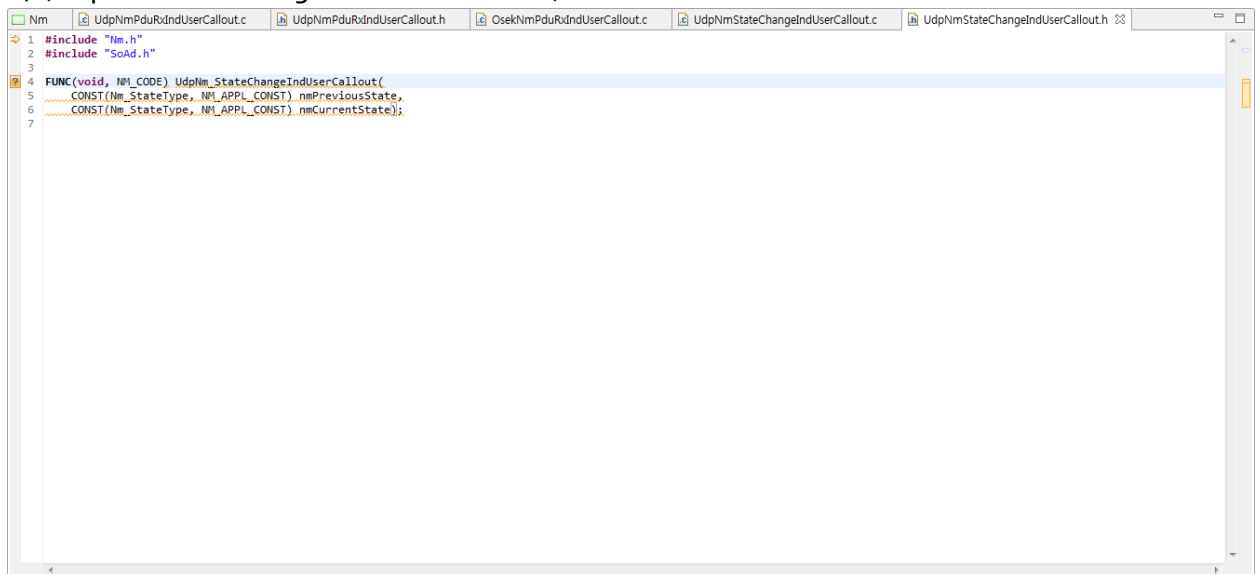
8.2.1.1 UdpNm 의 State 가 ReadySleep, PrepareBusSleep 일 경우에 Routing 을 Disable 하는 예제

Application 에서는 사용하고자 하는 용도와 각자 제어기 사양에 맞추어 직접 Code 를 작성해야 하며 리턴값 처리 및 예외 처리를 포함하여야 한다.

아래 Code 는 Sample 일 뿐으로 플랫폼에서는 아래 코드를 그대로 사용함으로써 발생하는 문제에 대해서 책임지지 않는다.

ComplexDeviceDriverSwComponent 를 추가하여 UserCallout Header 및 Source File 을 포함시켜 Component Level 로 올려서 사용할 수도 있다.

- 1) StateChangeIndUserCallout 을 위한 Header File 을 생성한다.(UdpNmStateChangeIndUserCallout.h)

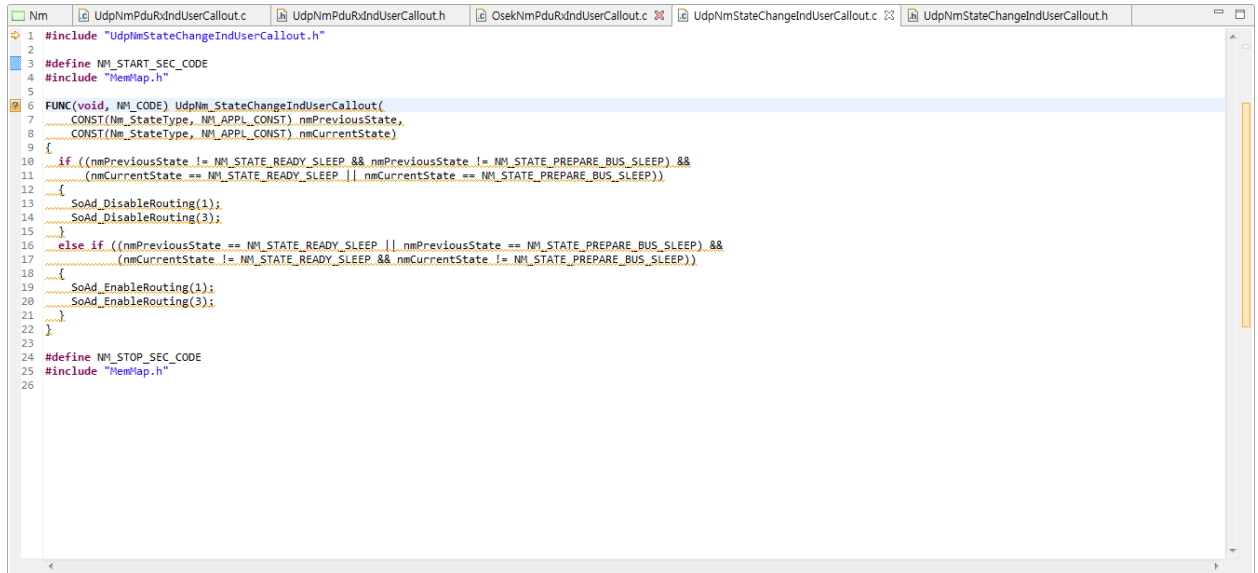


```

1 #include "Nm.h"
2 #include "SoAd.h"
3
4 FUNC(void, NM_CODE) UdpNm_StateChangeIndUserCallout(
5     CONST(Nm_StateType, NPL_APPL_CONST) nmPreviousState,
6     CONST(Nm_StateType, NPL_APPL_CONST) nmCurrentState);
7

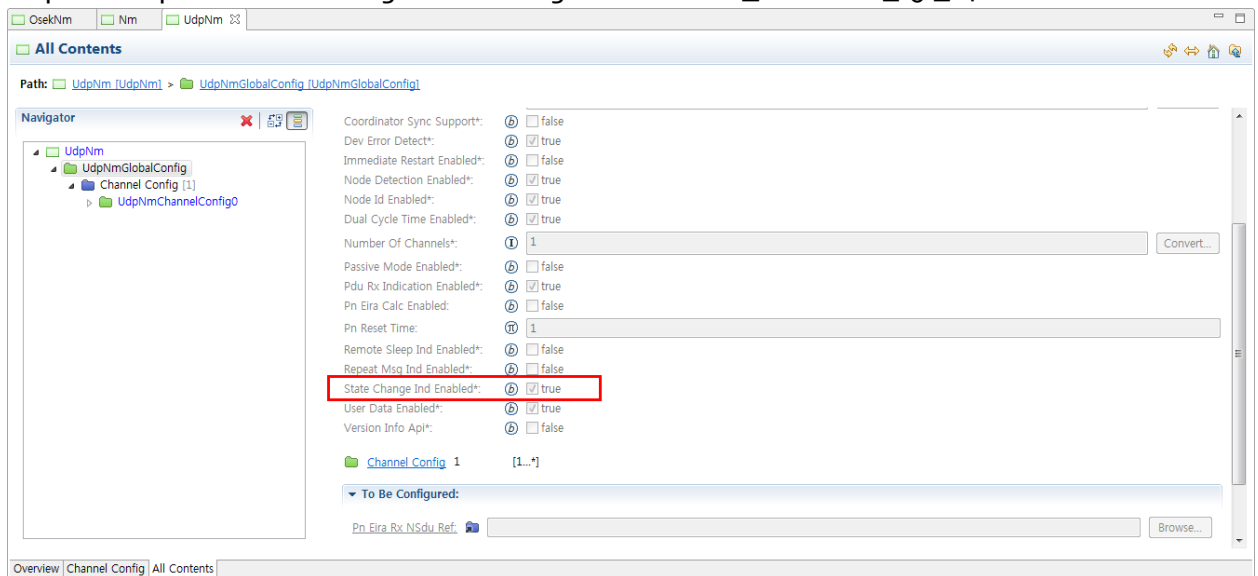
```

- 2) StateChangeIndUserCallout 을 위한 Source File 을 생성한다.(UdpNmStateChangeIndUserCallout.c)

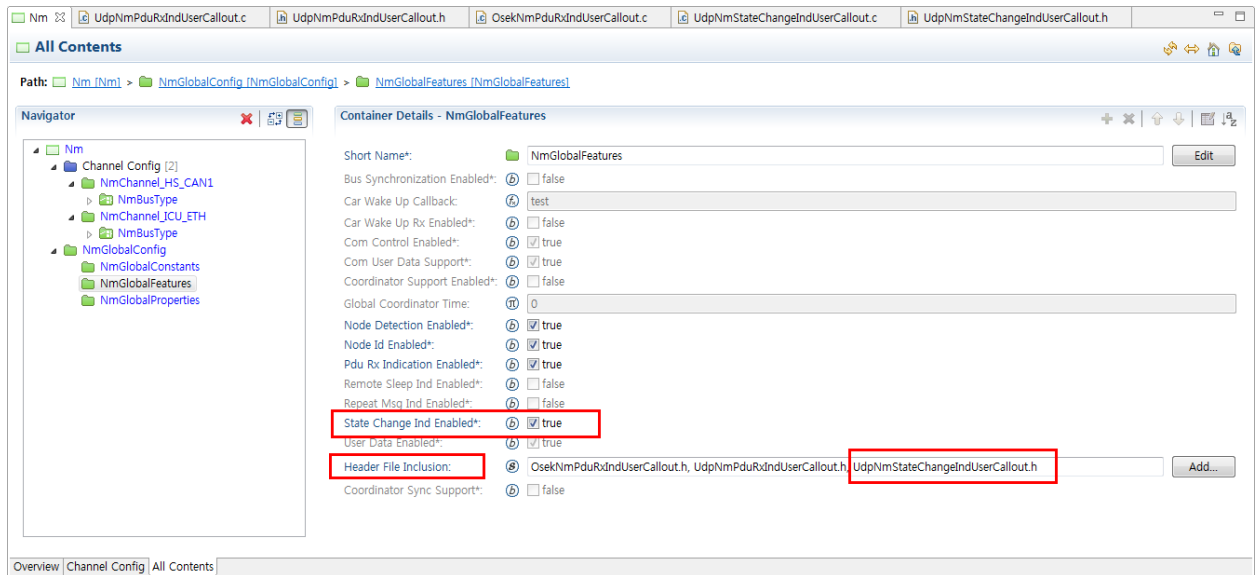


```
1 #include "UdpNmStateChangeIndUserCallout.h"
2
3 #define NM_START_SEC_CODE
4 #include "MemMap.h"
5
6 FUNC(void, NM_CODE) UdpNm_StateChangeIndUserCallout(
7     CONST(Nm_StateType, NM_APPL_CONST) nmPreviousState,
8     CONST(Nm_StateType, NM_APPL_CONST) nmCurrentState)
9 {
10     if ((nmPreviousState != NM_STATE_READY_SLEEP && nmPreviousState != NM_STATE_PREPARE_BUS_SLEEP) &&
11         (nmCurrentState == NM_STATE_READY_SLEEP || nmCurrentState == NM_STATE_PREPARE_BUS_SLEEP))
12     {
13         SoAd_DisableRouting(1);
14         SoAd_DisableRouting(3);
15     }
16     else if ((nmPreviousState == NM_STATE_READY_SLEEP || nmPreviousState == NM_STATE_PREPARE_BUS_SLEEP) &&
17             (nmCurrentState != NM_STATE_READY_SLEEP && nmCurrentState != NM_STATE_PREPARE_BUS_SLEEP))
18     {
19         SoAd_EnableRouting(1);
20         SoAd_EnableRouting(3);
21     }
22 }
23
24 #define NM_STOP_SEC_CODE
25 #include "MemMap.h"
26
```

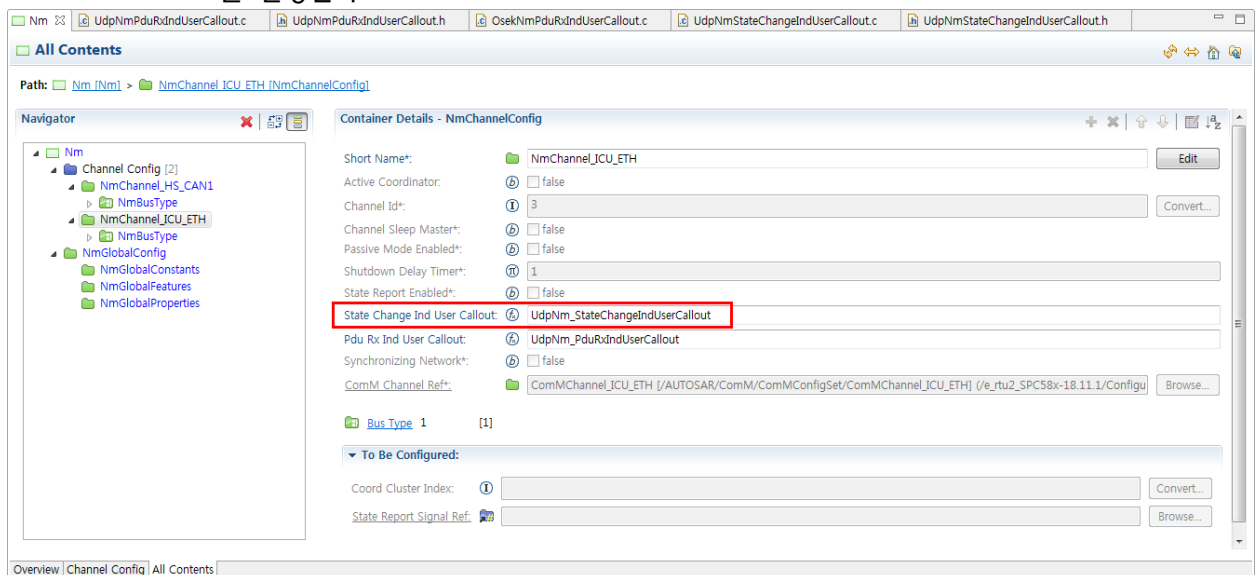
- 3) UdpNm/UdpNmGlobalConfig/StateChangeIndEnabled 를 true 로 설정한다.



- 4) Nm/GlobalConfig/NmGlobalFeatures/StateChangeIndEnabled 를 true 로 설정한다.
Nm/GlobalConfig/NmGlobalFeatures/HeaderFileInclusion 에 StateChangeIndUserCallout 을 위
한 Header File 을 설정한다.



- 5) Nm/NmChannelConfig/StateChangeIndUserCallout 에 StateChangeIndUserCallout 을 위한 Function Name 을 설정한다.



8.2.2 수신한 Nm Pdu Data 및 Node Identifier 를 알고 싶을 경우

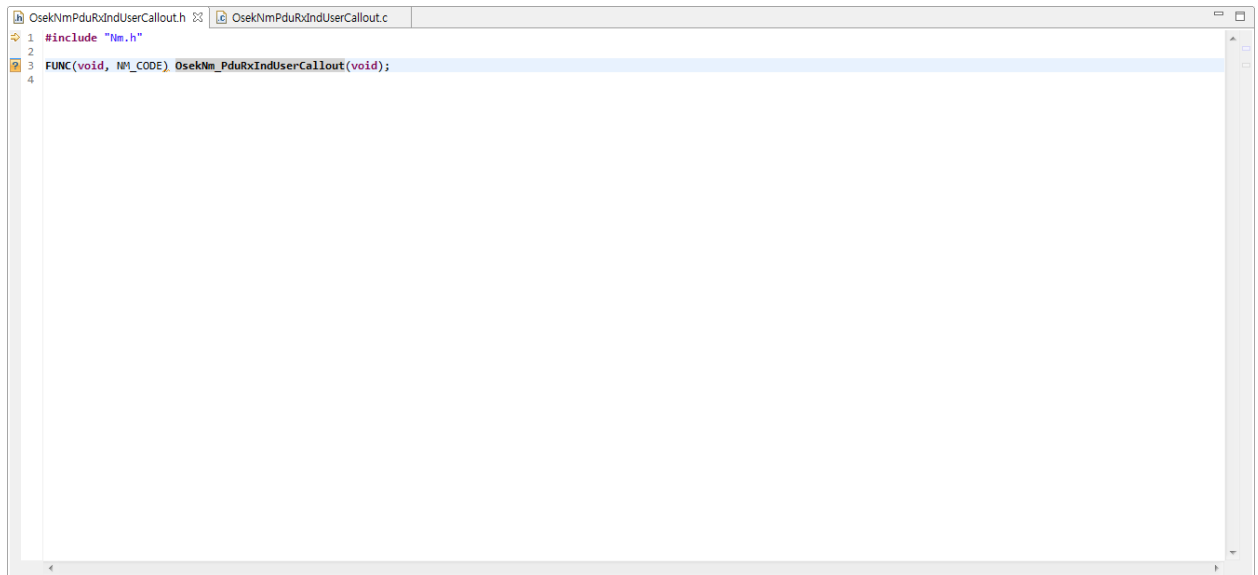
8.2.2.1 OsekNm 의 Nm Pdu 수신시 Pdu Data 및 Node Identifier 를 알고자 할 경우에 대한 예제

Application 에서는 사용하고자 하는 용도와 각자 제어기 사양에 맞추어 직접 Code 를 작성해야 하며 리턴값 처리 및 예외 처리를 포함하여야 한다.

아래 Code 는 Sample 일 뿐으로 플랫폼에서는 아래 코드를 그대로 사용함으로써 발생하는 문제에 대해서 책임지지 않는다.

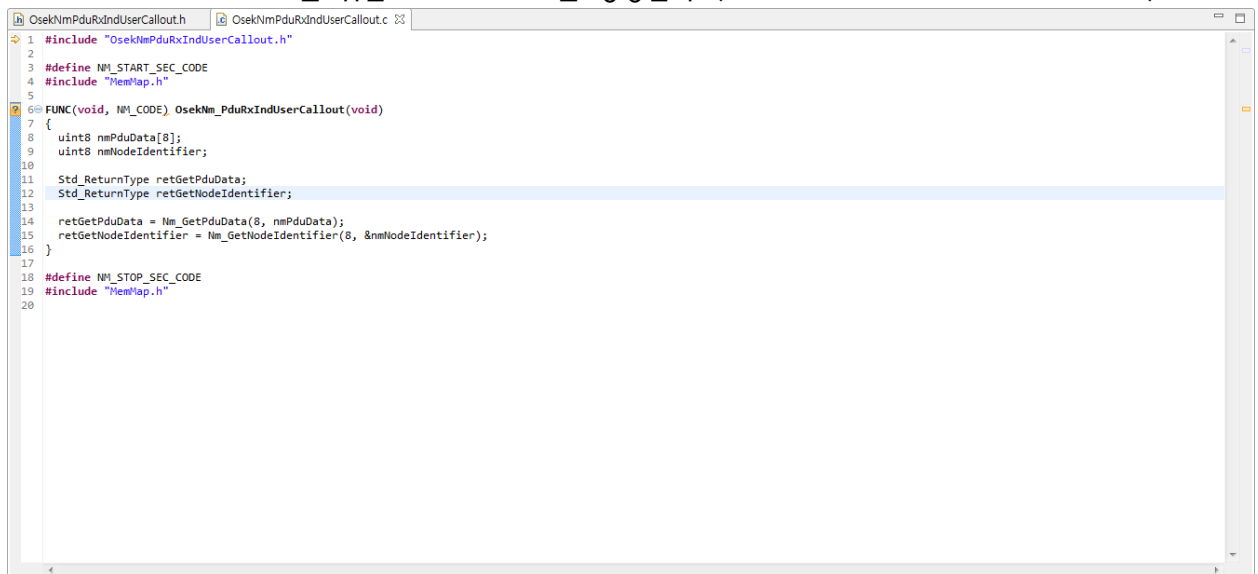
ComplexDeviceDriverSwComponent 를 추가하여 UserCallout Header 및 Source File 을 포함시켜 Component Level 로 올려서 사용할 수도 있다.

- 1) PduRxIndUserCallout 을 위한 Header File 을 생성한다.(OsekNmPduRxIndUserCallout.h)



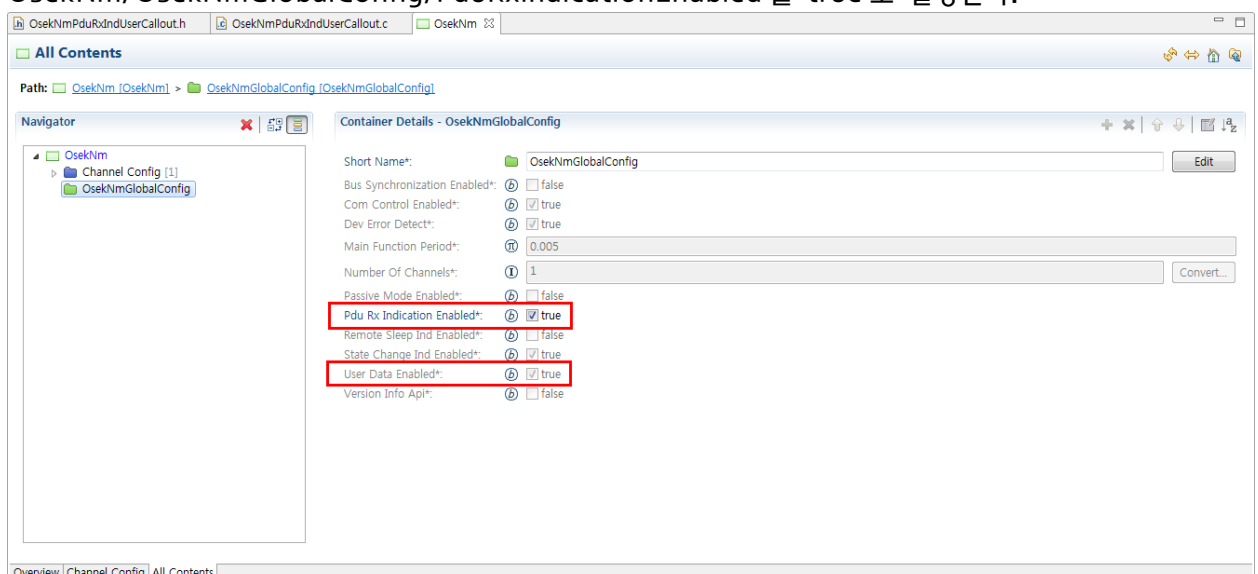
```
OsekNmPduRxIndUserCallout.h OsekNmPduRxIndUserCallout.c
1 #include "Nm.h"
2
3 FUNC(void, NM_CODE) OsekNm_PduRxIndUserCallout(void);
4
```

2) PduRxIndUserCallout 을 위한 Source File 을 생성한다. (OsekNmPduRxIndUserCallout.c)

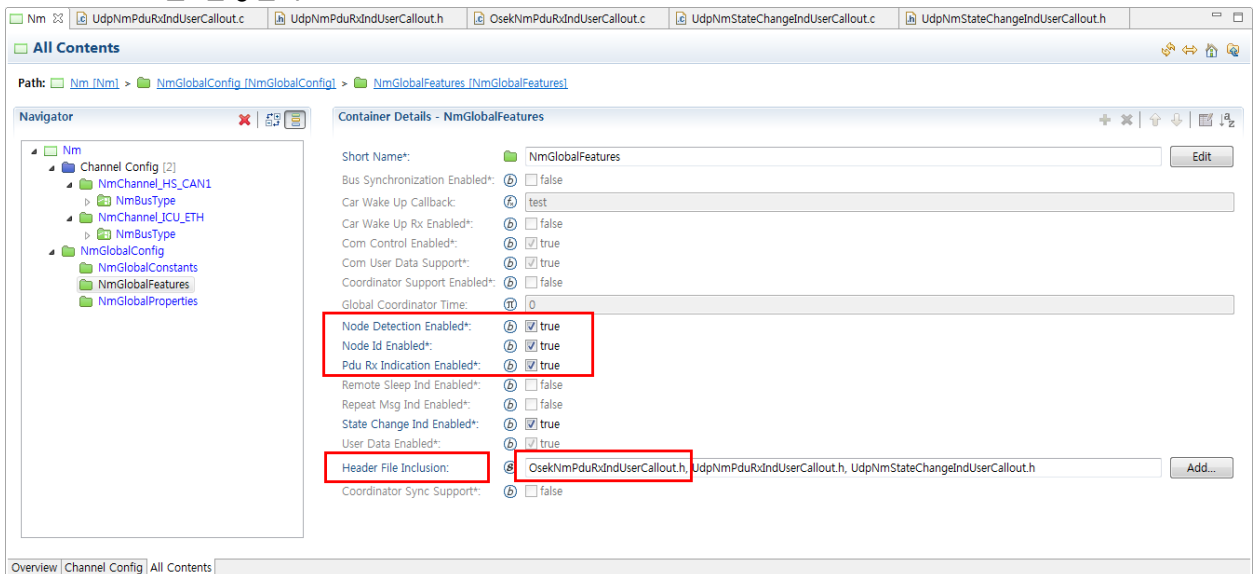


```
OsekNmPduRxIndUserCallout.h OsekNmPduRxIndUserCallout.c
1 #include "OsekNmPduRxIndUserCallout.h"
2
3 #define NM_START_SEC_CODE
4 #include "NmMap.h"
5
6 FUNC(void, NM_CODE) OsekNm_PduRxIndUserCallout(void)
7 {
8     uint8 nmPduData[8];
9     uint8 nmNodeIdentifier;
10
11     Std_ReturnType retGetPduData;
12     Std_ReturnType retGetNodeIdentifier;
13
14     retGetPduData = Nm_GetPduData(8, nmPduData);
15     retGetNodeIdentifier = Nm_GetNodeIdentifier(8, &nmNodeIdentifier);
16 }
17
18 #define NM_STOP_SEC_CODE
19 #include "NmMap.h"
20
```

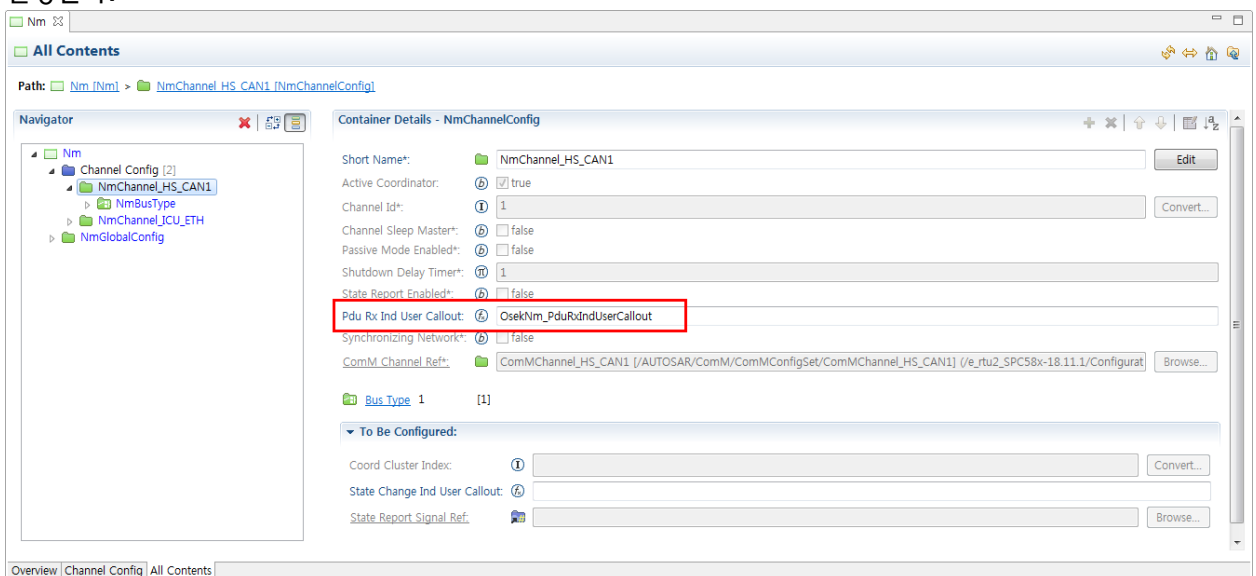
3) OsekNm/OsekNmGlobalConfig/PduRxIndicationEnabled 를 true 로 설정한다. OsekNm/OsekNmGlobalConfig/PduRxIndicationEnabled 를 true 로 설정한다.



- 4) Nm/NmGlobalConfig/NmGlobalFeatures/NodeDetectionEnabled 를 true 로 설정한다.
 Nm/NmGlobalConfig/NmGlobalFeatures/NodeIdEnabled 를 true 로 설정한다.
 Nm/NmGlobalConfig/NmGlobalFeatures/PduRxIndicationEnabled 를 true 로 설정한다.
 Nm/NmGlobalConfig/NmGlobalFeatures/HeaderFileInclusion 에 PduRxIndUserCallout 을 위한 Header File 을 설정한다.



- 5) Nm/NmChannelConfig/PduRxIndUserCallout 에 PduRxIndUserCallout 을 위한 Function Name 을 설정한다.



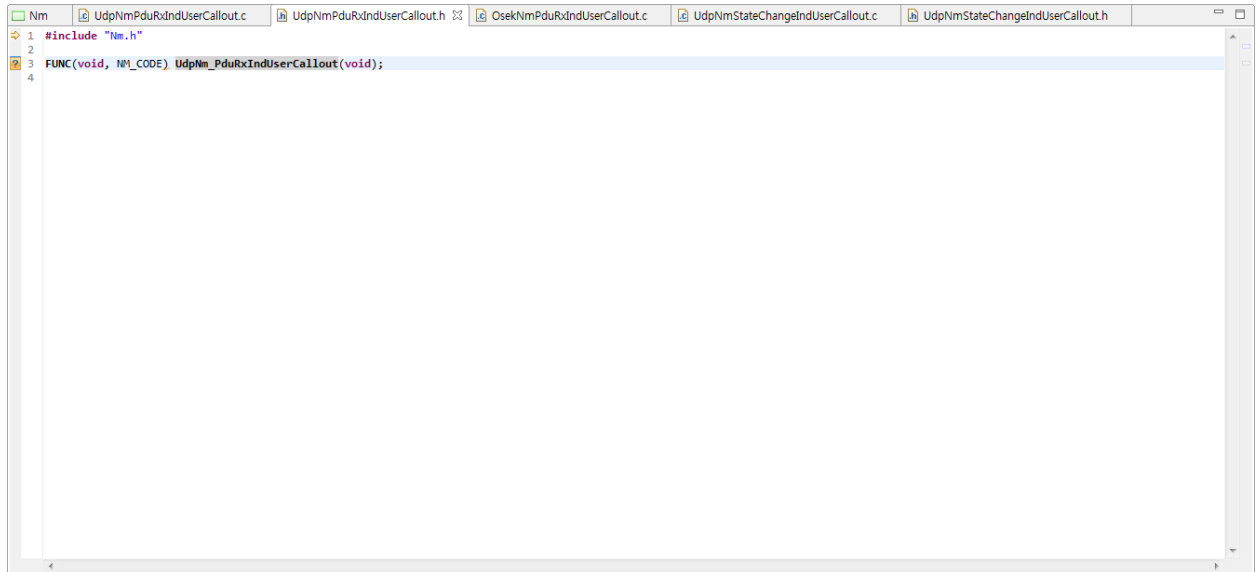
8.2.2.2 UdpNm 의 Nm Pdu 수신시 Pdu Data 및 Node Identifier 를 알고자 할 경우에 대한 예제

Application 에서는 사용하고자 하는 용도와 각자 제어기 사양에 맞추어 직접 Code 를 작성해야 하며 리턴값 처리 및 예외 처리를 포함하여야 한다.

아래 Code 는 Sample 일 뿐으로 플랫폼에서는 아래 코드를 그대로 사용함으로써 발생하는 문제에 대해서 책임지지 않는다.

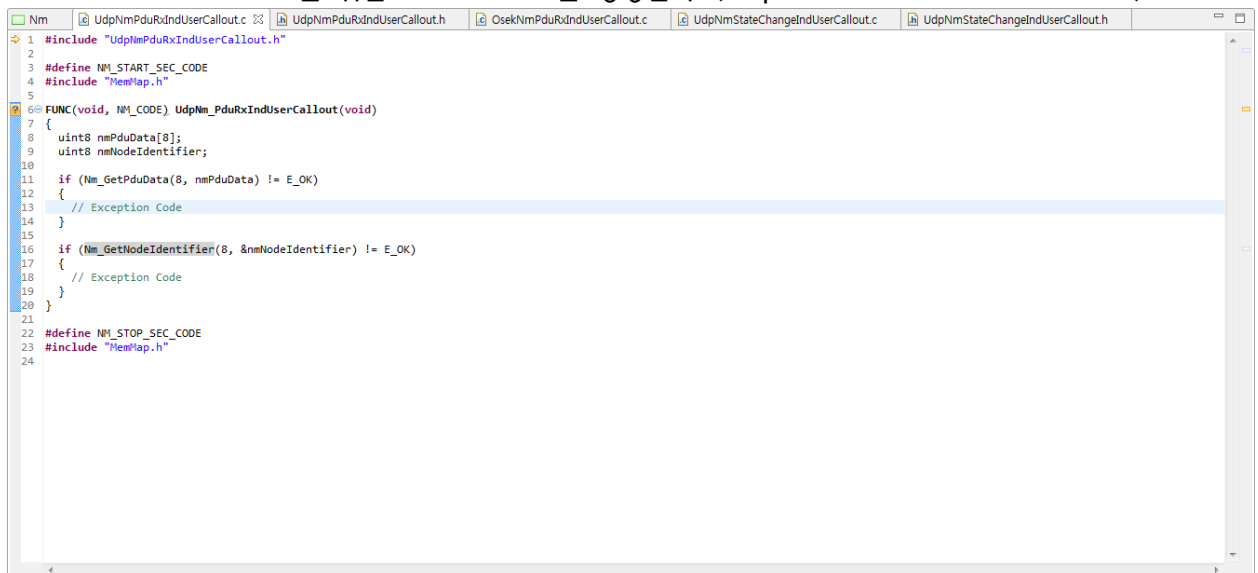
ComplexDeviceDriverSwComponent 를 추가하여 UserCallout Header 및 Source File 을 포함시켜 Component Level 로 올려서 사용할 수도 있다.

1) PduRxIndUserCallout 을 위한 Header File 을 생성한다.(UdpNmPduRxIndUserCallout.h)



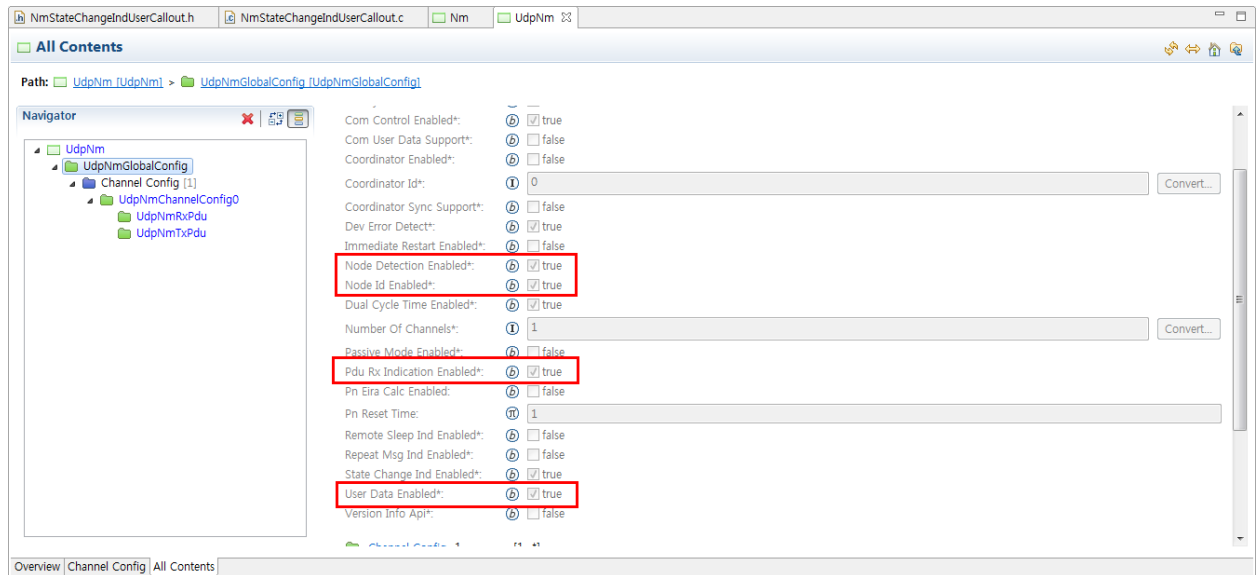
```
1 // Nm.h
2
3 FUNC(void, NM_CODE) UdpNm_PduRxIndUserCallout(void);
4
```

2) PduRxIndUserCallout 을 위한 Source File 을 생성한다. (UdpNmPduRxIndUserCallout.c)

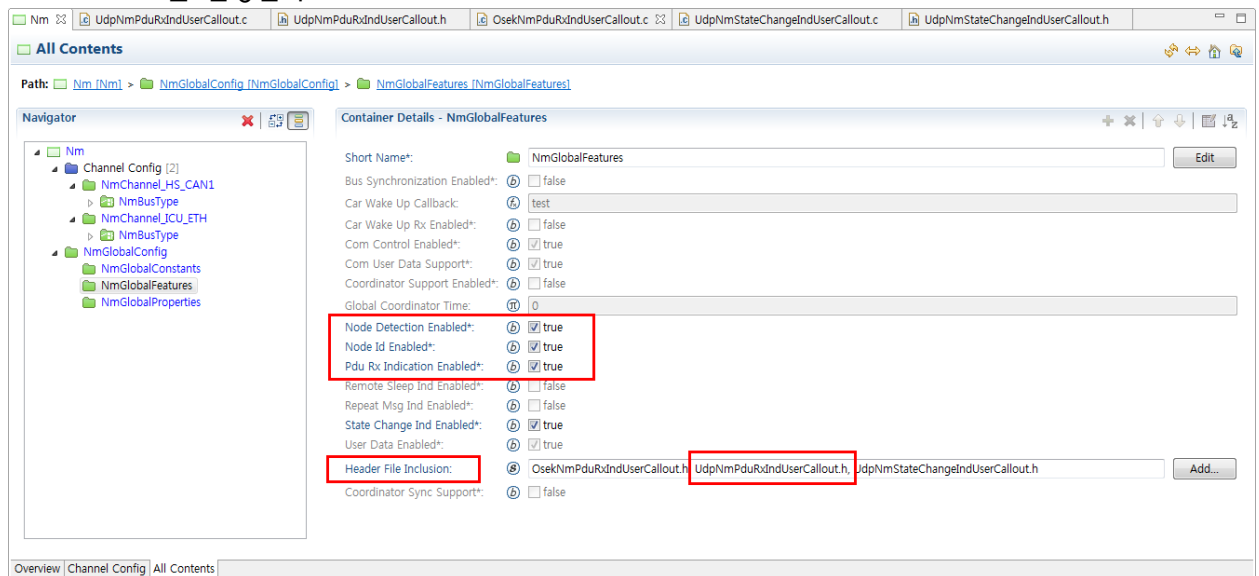


```
1 // UdpNmPduRxIndUserCallout.h
2
3 #define NM_START_SEC_CODE
4 #include "NmMap.h"
5
6 FUNC(void, NM_CODE) UdpNm_PduRxIndUserCallout(void)
7 {
8     uint8 nmPduData[8];
9     uint8 nmNodeIdentifier;
10
11     if (Nm_GetPduData(8, nmPduData) != E_OK)
12     {
13         // Exception Code
14     }
15
16     if (Nm_GetNodeIdentifier(8, &nmNodeIdentifier) != E_OK)
17     {
18         // Exception Code
19     }
20 }
21
22 #define NM_STOP_SEC_CODE
23 #include "NmMap.h"
24
```

- 3) UdpNm/UdpNmGlobalConfig/NodeDetectionEnabled 를 true 로 설정한다.
UdpNm/UdpNmGlobalConfig/NodeIdEnabled 를 true 로 설정한다.
UdpNm/UdpNmGlobalConfig/PduRxIndicationEnabled 를 true 로 설정한다.
UdpNm/UdpNmGlobalConfig/UserDataEnabled 를 true 로 설정한다.



- 4) Nm/NmGlobalConfig/NmGlobalFeatures/NodeDetectionEnabled 를 true 로 설정한다.
 Nm/NmGlobalConfig/NmGlobalFeatures/NodeIdEnabled 를 true 로 설정한다.
 Nm/NmGlobalConfig/NmGlobalFeatures/PduRxIndicationEnabled 를 true 로 설정한다.
 Nm/NmGlobalConfig/NmGlobalFeatures/HeaderFileInclusion 에 PduRxIndUserCallout 을 위한 Header File 을 설정한다.



- 5) Nm/NmChannelConfig/PduRxIndUserCallout 에 PduRxIndUserCallout 을 위한 Function Name 을 설정한다.

